Proposed Solution: Multi-Tier Modern Data Analytics Pipeline

Phase 1: Solution Architecture & Technology Stack

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1. Multi-Tier Solution Blueprint

The proposed solution adopts a four-tier enterprise architecture separating data ingestion, processing, business logic, and presentation. Raw datasets are cleaned using custom vectorized pipelines in Python, generating normalized datasets optimized for both SQL-style relational queries and Tableau visual calculations.

- Tier 1: Data Ingestion & Schema Normalization (clean_data.py)
- Tier 2: Analytical Core & Econometric Modeling (analyze.py, exploratory_analysis.py)
- Tier 3: Visual Analytics & Interactive Dashboards (Tableau Public)
- Tier 4: Enterprise Presentation & REST API Delivery (Olympic-Flask Web Portal)